EXPLORATORY RESEARCH REPORT

# Predicting Answer Correctness in a Small Language Model

*Semantic entropy, token predictive entropy and hidden-state dispersion on Qwen3-1.7B and TriviaQA*

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## Abstract

Language models often produce fluent answers that are wrong. This report asks whether uncertainty signals computed from a model's own outputs and internal states can predict whether a separate, low-temperature answer to the same question is correct. Using Qwen3-1.7B (4-bit) on 100 TriviaQA questions, ten high-temperature answers per question are used to compute three kinds of signal: empirical semantic entropy over NLI-based meaning clusters, token predictive entropy from teacher-forced next-token distributions, and within-question dispersion of hidden-state representations.

Both entropy signals rank incorrect answers well (AUROC 0.841 for semantic entropy and 0.852 for token entropy) and clearly beat random rejection (AURAC 0.463 and 0.538 against a random baseline of about 0.270). Paired bootstrap tests cannot separate the two. Adding predictor-state hidden-state dispersion to a logistic-regression probe gives the strongest exploratory result (AUROC 0.839, AURAC 0.515, F1 0.553), but its gain over entropy-only features is not statistically established. All results come from one model, one dataset slice and 100 questions, and are reported as a pilot rather than as a validated hallucination detector.

## Key findings

- **Entropy signals carry real information.** On 100 questions, semantic entropy and token entropy reach AUROC 0.841 and 0.852 for flagging incorrect answers. Rejecting the most uncertain questions first lifts retained accuracy well above the 0.27 base rate.
- **Neither entropy signal is shown to be better.** Token entropy has higher point estimates, but every paired 95% interval for token minus semantic entropy includes zero.
- **Hidden states alone were weaker than entropy.** The best hidden-only probe (layer index 28, mean pooling) reached AUROC 0.772, below the entropy baselines.
- **Hidden-state dispersion is promising but inconclusive.** Adding predictor-state dispersion ($p-1$) to the entropy features gave the best numerical results of any probe tested, but all primary paired intervals include zero.
- **Scope is narrow.** One model, one dataset slice, 100 questions, 27 correct labels, and an exact-match correctness proxy. No generalization claim is made.

## 1. Introduction

The central research question is:

*Can information extracted from a language model's outputs and internal representations help predict whether its answer is correct?*

Three families of signal are studied, each measuring something different:

1. **Semantic uncertainty.** Do repeated answers to the same question differ in meaning?
2. **Token predictive uncertainty.** How spread out is the model's next-token distribution along its answers?
3. **Hidden-state geometry.** Do the model's internal vectors vary across answers in a way that relates to correctness?

The work adapts the semantic-entropy approach of Farquhar et al. (2024) to a small open model. It is an experimental extension, not an exact reproduction of every step of the original paper. In particular, the semantic entropy used here is an empirical cluster-frequency entropy rather than the probability-weighted version (see Section 3.1).

# 2. Experimental design

| Item | Setting |
|---|---|
| Base model | Qwen3-1.7B, 4-bit NF4 quantization, double quantization, FP16 compute |
| Dataset | TriviaQA (unfiltered.nocontext), 100 validation questions |
| Feature-generating answers | 10 per question, 1,000 in total; T = 1.0, top-p = 0.8, top-k = 20 |
| Target answer | 1 separate answer per question; T = 0.1 |
| Prompting | Five few-shot examples; max_new_tokens = 64 |
| Correctness label | Normalized reference-based exact match (lowercase, strip ASCII punctuation, remove articles, collapse whitespace) |
| Label balance | 27 correct, 73 incorrect (base accuracy 0.27) |
| Model dimensions | Hidden size 2,048; 28 decoder blocks; 29 returned hidden-state indices |
| NLI model | cross-encoder/nli-deberta-v3-small |

**Cross-temperature design.** Signals are computed from the ten T = 1.0 answers and used to predict the correctness of a separate T = 0.1 answer. This is deliberately different from asking whether each of the 1,000 sampled answers is correct.

**Statistical unit.** The unit is the question. There are 100 question-level targets, not 1,000 independent ones, because the ten answers to a question share one prompt and one label. All bootstrap resampling is therefore done at the question level.

# 3. Methods

## 3.1 Empirical semantic entropy

For each question, the ten answers are grouped by meaning:

- Every unordered pair (45 pairs) is tested in both directions with the NLI model, using question-conditioned premise and hypothesis strings. This gives 90 directional calls per question and 9,000 in total.
- A pair is mutually entailing only if both directions predict entailment with score ≥ 0.5. The 0.5 threshold is a chosen criterion, not an established standard.
- Answers are assigned by greedy complete-linkage clustering: an answer joins the first cluster in which it matches every existing member.

With cluster sizes n(i) over N = 10 answers, p(i) = n(i) / 10 and the entropy is $H = -\Sigma\, p(i) \ln p(i)$, measured in nats.

**Important distinction.** This is empirical cluster-frequency entropy. The probability-weighted version instead sums model probabilities of the answers inside each cluster. Generation-time log-probabilities and the filtered sampling distribution were not saved, so the weighted version could not be recovered exactly. A teacher-forced, length-normalized approximation was investigated as an auxiliary analysis only.

## 3.2 Token predictive entropy

Saved answers are rebuilt as prompt + answer and teacher-forced through Qwen3. For an answer token at position p, the next-token distribution is taken from the causal predictor at position p−1, using the raw full-vocabulary softmax:

$$H(t) = -\Sigma\, P(v \mid \text{context}) \ln P(v \mid \text{context})$$

The question-level score is the mean of the ten answer-level mean token entropies. Generation-time top-k and top-p filtering is not reproduced. Tokens that cross the prompt/answer character boundary (347 of 1,000 answers) are kept in the primary analysis, and a boundary-exclusion variant is reported as sensitivity analysis. End-of-sequence tokens are not appended or scored.

## 3.3 Hidden-state features and dispersion

Hidden states are extracted for all 29 indices (index 0 is the token-embedding output, indices 1–27 are decoder block outputs, and index 28 is the final RMSNorm output) under four pooling strategies (first, last, mean, max). Two alignments are kept as separate measurements:

- **Predictor state (p−1):** the state used to predict each answer token, extracted for all 29 indices and four poolings.
- **Post-token state (p):** the state after the answer token has been incorporated, extracted separately for index 28 with mean pooling.

For a fixed layer and pooling method, the question-level mean representation is $Z(q) = (1/10)\, \Sigma\, z(q, r)$, and the within-question dispersion is:

$$D(q) = (1/10)\, \Sigma\, \| z(q, r) - Z(q) \|_2$$

Dispersion is computed in float64 from the ten answer vectors. It measures variation in representation space; it is not itself a semantic-disagreement score.

## 3.4 Evaluation protocols

Two protocols are used and their numbers are never mixed. **Direct raw-score evaluation** ranks the signal values themselves, with higher uncertainty meaning a stronger candidate for rejection; no model is trained. **Predictive-probe evaluation** trains logistic regression and scores out-of-fold probabilities, with uncertainty defined as 1 − P(correct).

Metrics:

- **Spearman correlation** against the correctness label (negative values mean higher uncertainty goes with lower correctness).
- **AUROC** with incorrectness as the positive class.
- **AURAC** (area under the retained-accuracy versus coverage curve): reject the most uncertain question first, break exact ties by ascending question ID, integrate with the trapezoidal rule, and use the one-retained-question accuracy at zero coverage.
- **Question-level bootstrap** percentile intervals (95%).

| Evaluation | Resamples | Seed | What is resampled |
|---|---|---|---|
| Raw-score evaluation | 20,000 | 20261031 | Paired question-level resamples; metrics recomputed |
| Initial 235-probe screening | 100 | 20261106 | Saved out-of-fold predictions (preliminary) |
| Dispersion probe comparisons | 20,000 | 20261106 | Paired resamples of saved out-of-fold predictions |

Probe intervals are conditional on the saved out-of-fold predictions; models are not refit in each replicate.

### 3.5 Predictive probes

Probes use two-fold stratified nested cross-validation at the question level (shuffle on, random_state 20261106), with StandardScaler inside the training pipeline and logistic regression (lbfgs, max_iter 1,000). The regularization strength $C \in \{0.1, 1.0\}$ is chosen by inner-fold AUROC. The initial screening covered 235 configurations: 116 hidden-only (29 indices × 4 poolings), 3 entropy baselines, and 116 hidden-plus-entropy.

## 4. Results

### 4.1 Raw entropy and dispersion scores

| Raw signal | Spearman | AUROC (incorrectness) | AURAC |
|---|---|---|---|
| Empirical semantic entropy | −0.528 | 0.841 | 0.463 |
| Token predictive entropy | −0.541 | 0.852 | 0.538 |
| Post-token dispersion (index 28, mean) | −0.397 | 0.758 | 0.436 |
| Random rejection (reference) | n/a | 0.500 | 0.270 |

95% question-level bootstrap intervals (20,000 resamples):

| Signal | Spearman | AUROC | AURAC |
|---|---|---|---|
| Semantic entropy | [−0.652, −0.382] | [0.754, 0.914] | [0.326, 0.605] |
| Token entropy | [−0.678, −0.372] | [0.749, 0.936] | [0.403, 0.653] |
| Post-token dispersion | [−0.560, −0.208] | [0.638, 0.863] | [0.297, 0.575] |

The random-rejection AURAC was simulated at a mean of 0.2696 (95% interval 0.187 to 0.361), consistent with the overall accuracy of 0.27. The intervals are wide, as expected with 100 questions.

### 4.2 Semantic versus token entropy

| Difference (token − semantic) | Point difference | 95% paired interval |
|---|---|---|
| Spearman | −0.013 | [−0.109, 0.085] |
| AUROC | +0.010 | [−0.055, 0.075] |
| AURAC | +0.075 | [−0.003, 0.142] |

All three intervals include zero. Token entropy has higher point estimates for AUROC and AURAC, but statistical superiority is not established. This does not show the two are equal; it shows this sample cannot tell them apart.

### 4.3 Robustness checks

**Clustering stability.** Answer order was shuffled 1,000 times per question (100,000 runs). Partitions changed for 7 of 100 questions (2.7% of runs) and entropy varied for 5 questions, with a mean range of 0.008 nats and a maximum of 0.348 nats. Greedy clustering is stable for most questions but not

perfectly order-independent.

**Alternative clustering.** A connected-components alternative, which can over-merge through transitive links, gave AUROC 0.838 versus 0.842 for greedy complete linkage and differed on 7 questions. Greedy complete linkage was kept as the primary method.

**Manual validation of the NLI matching.** On 44 unique answer pairs from 37 questions (one annotator): 19 true positives, 1 false positive, 1 false negative, 23 true negatives, giving accuracy 95.5% and precision, recall and F1 of 0.95. This is a small check, not a definitive human benchmark.

**Token entropy checks.** All 100 questions and 1,000 answers were scored with no failures, and entropy values were finite and non-negative. An optimization test matched a full-logit calculation within $2.4 \times 10^{-6}$ nats (tolerance $10^{-5}$), though that test covered one real answer. Excluding the first token of boundary-crossing answers lowered mean entropy from 0.484 to 0.452 and gave a rank correlation of 0.706 with the primary scores, so the boundary convention matters for some questions.

## 4.4 Hidden-state geometry

Mean-pooled representation size grows through the decoder blocks and drops sharply at index 28, consistent with the final RMSNorm. A larger norm does not mean a better representation.

| Hidden-state index | Mean L2 norm | Mean within-question dispersion |
|---|---|---|
| 0 (embeddings) | 0.95 | 0.43 |
| 5 | 31.12 | 9.17 |
| 15 | 144.00 | 29.63 |
| 25 | 1,551.20 | 432.29 |
| 27 | 2,241.93 | 562.53 |
| 28 (final RMSNorm) | 90.68 | 27.65 |

For index 28 with mean pooling, dispersion has mean 27.65 for predictor states (p−1) and 29.45 for post-token states (p). Some questions have exactly zero dispersion (19 for p−1 in the original float32 analysis, 15 for p in a float64 recomputation). Controlled checks showed this is expected: when answers differ only at their final token, their p−1 context states are identical, whereas p states can tell them apart. An identical-answer control stayed identical at both alignments, so this is a causal-position effect, not an indexing bug.

## 4.5 Exploratory probe screening (235 configurations)

| Probe | AUROC | AURAC | Spearman |
|---|---|---|---|
| Semantic entropy only | 0.838 | 0.480 | −0.521 |
| Token entropy only | 0.802 | 0.479 | −0.465 |
| Semantic + token entropy | 0.822 | 0.495 | −0.495 |
| Best hidden-only (index 28, mean) | 0.772 | 0.463 | −0.418 |
| Best hidden + entropy by AUROC (index 28, mean) | 0.805 | 0.482 | −0.469 |

The hidden-only and hidden-plus-entropy leaders by AUROC were both below the semantic-entropy probe. The AURAC leaders were different configurations (index 25 with last pooling, and index 22 with last pooling plus entropy), so no single best hidden-state model should be claimed without naming the metric. Searching 116 alternatives introduces selection optimism that the preliminary bootstrap does not correct.

At the fixed 0.5 threshold, the semantic-only and token-only probes predicted no correct cases (F1 = 0, accuracy equal to the 0.73 majority class). This is why ranking and thresholded metrics are reported separately.

### 4.6 Does dispersion add information beyond entropy?

Two dispersion features were compared against the semantic + token entropy probe (A), using identical outer and inner splits.

| Configuration | AUROC | AURAC | Spearman | Acc. | Prec. | Recall | F1 |
|---|---|---|---|---|---|---|---|
| A. Semantic + token entropy | 0.822 | 0.495 | −0.495 | 0.77 | 0.750 | 0.222 | 0.343 |
| B. A + predictor-state dispersion (p−1) | 0.839 | 0.515 | −0.522 | 0.79 | 0.650 | 0.481 | 0.553 |
| C. A + post-token dispersion (p) | 0.832 | 0.509 | −0.511 | 0.77 | 0.611 | 0.407 | 0.489 |

Paired comparisons (20,000 resamples):

| Comparison | AUROC diff. | 95% interval | AURAC diff. | 95% interval |
|---|---|---|---|---|
| B − A (p−1 over entropy) | +0.017 | [−0.028, 0.069] | +0.020 | [−0.021, 0.068] |
| C − A (p over entropy) | +0.010 | [−0.034, 0.058] | +0.015 | [−0.027, 0.061] |
| C − B (p versus p−1) | −0.007 | [−0.023, 0.008] | −0.006 | [−0.021, 0.004] |

Predictor-state dispersion has the strongest numerical results, and it raised recall at the 0.5 threshold from 0.222 to 0.481. However, every primary AUROC and AURAC interval includes zero. The recall gain (interval 0.038 to 0.48 for B − A) excludes zero, but recall is threshold-dependent and was one of several metrics inspected without multiplicity correction, so it is treated as a secondary observation.

## 5. Discussion

**What the results support.** On this sample, uncertainty computed from ten sampled answers is a useful ranking signal for whether a separate low-temperature answer is correct. Both entropy signals exceed random rejection by a wide margin, and a semantic-entropy-only probe already reaches AUROC 0.838.

**What they do not support.** The data do not show that token entropy beats semantic entropy, that hidden states predict correctness better than entropy, or that dispersion adds value beyond entropy. Hidden-state dispersion improves the combined probe only numerically, and part of that improvement may reflect the search over many configurations.

**Why the hidden-state result is still interesting.** Dispersion is cheap once hidden states are available, and predictor-state dispersion produced the best point estimates under identical splits. A properly powered, pre-specified test is the natural next step.

**Framing.** This is an experimental study of cross-temperature correctness prediction. It does not establish a calibrated, end-to-end hallucination detector, causality, or generalization to unseen datasets and model families.

## 6. Limitations

| Limitation | Implication |
|---|---|
| 100 questions, 27 correct labels | Small sample and class imbalance; estimates and fold selection can be unstable |
| One model, one dataset slice | No cross-model or cross-dataset generalization |
| Exact-match correctness | A proxy, not human-verified truth; valid paraphrases can be marked wrong |
| Cross-temperature target | Features come from T = 1.0 answers; the label belongs to a separate T = 0.1 answer |
| Teacher-forced reconstruction | Token distributions and hidden states are rebuilt from saved text, not captured during generation |

| Limitation | Implication |
|---|---|
| Full-vocabulary probabilities | Generation-time top-k and top-p filtering is not recovered |
| Empirical semantic entropy | Not the probability-weighted estimator of the original paper |
| 235-configuration screening | Best-of-many selection optimism, no multiple-comparison adjustment |
| Bootstrap on saved predictions | Intervals do not refit models or capture all selection variability |
| Provenance gaps | Checkpoint hash and full environment lock were not recorded for the historical run |

## 7. Next steps

1. Expand the question sample and use a more robust correctness target, including human review of ambiguous or paraphrased answers.
2. Validate on an independent question-answering dataset.
3. Evaluate another model checkpoint or model family.
4. Pre-specify a dispersion feature, or use a nested feature-selection procedure, instead of choosing the best of 116 on the full data.
5. Consider calibration and selective-risk analysis only after larger-scale validation.

## 8. Reproducibility

**Recorded:** model and quantization settings, dataset slice, generation settings, correctness normalization, tokenization alignment, hidden-state tensor shapes and dtypes (per-answer 1,000 × 29 × 4 × 2,048 in float16; question-level 100 × 29 × 4 × 2,048 in float32; post-token 1,000 × 2,048 in float32), AURAC conventions, bootstrap settings and seeds, and cross-validation settings.

**Open items:** exact checkpoint hash and revision; Python, PyTorch, Transformers, bitsandbytes, SciPy, scikit-learn and CUDA versions; hardware details; the generation seed and full chat-template provenance; the NLI label order, maximum sequence length and truncation count; and the seed and provenance of the random-rejection simulation. Some large generated artifacts (generation JSON, teacher-forced scoring files, hidden-state tensors) are local and not all committed to the public repository.

## References

Farquhar, S., Kossen, J., Kuhn, L., & Gal, Y. (2024). Detecting hallucinations in large language models using semantic entropy. *Nature*.

## Glossary

- **Nat:** unit of entropy when natural logarithms are used.
- **AUROC:** probability that a randomly chosen incorrect answer receives a higher uncertainty score than a randomly chosen correct one (0.5 is chance, 1.0 is perfect).
- **AURAC:** area under the curve of retained accuracy against coverage when the most uncertain questions are rejected first; higher means rejecting uncertain questions helps more.
- **Teacher forcing:** feeding a saved answer back through the model to read its internal predictions, rather than generating it fresh.

- **p−1 and p:** the hidden state used to predict an answer token, and the hidden state after that token is incorporated.
- **Out-of-fold prediction:** a prediction made by a model that did not see that question during training.